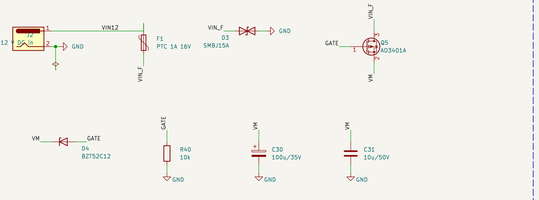
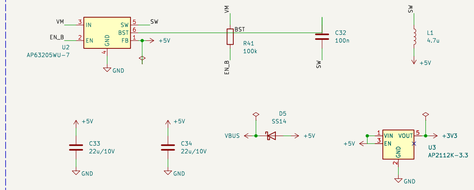
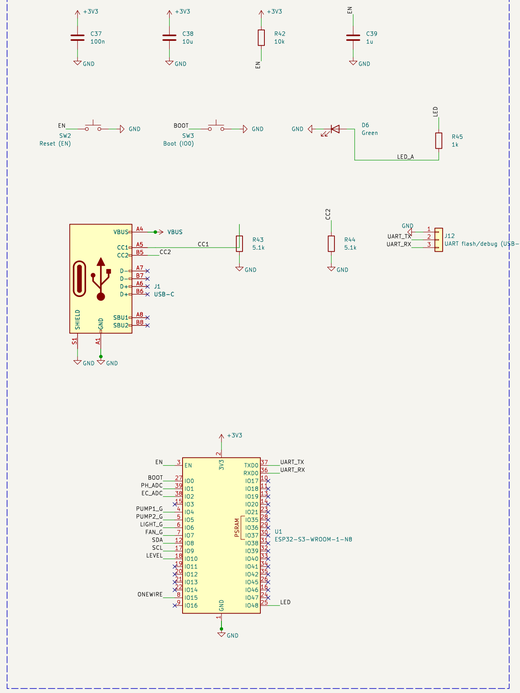
POWER IN: 12 V, PTC, TVS, reverse-polarity PMOS with 12 V gate clamp, bulk



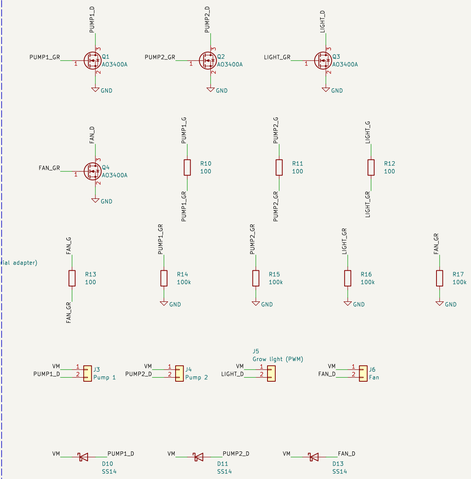
5 V BUCK AP63205 + 3.3 V LDO



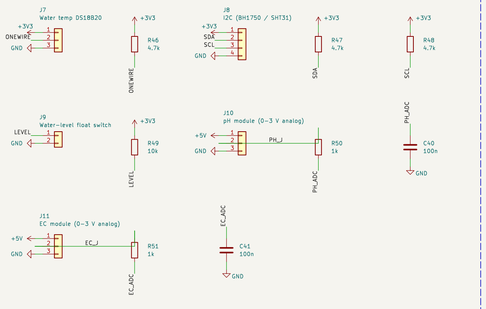
CONTROLLER: ESP32-S3 (WiFi + MQTT), USB-C



LOW-SIDE LOAD DRIVERS: 2 pumps, grow-light PWM, fan (12 V)



SENSORS: water temp, I2C, level switch, pH and EC analog inputs



BioTray: ESP32-S3 grows-tray controller. Two 12 V pumps and a fan are switched low-side with flyback diodes; the grow light is PWM dimmed (no flyback diode; LED strip). Pump, fan and light total must stay under the PTC rating (1.1 A hold): fit a larger PTC or fuse for bigger loads. pH and EC are off-the-shelf analog modules with 0–3 V outputs; calibrate in firmware. Water-level float switch closes to GND when water is present.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

Sheet: /

File: biotray.kicad_sch

Title: BioTray vertical-farm tray controller (ESP32-S3, MQTT)

Size: A1	Date: 2026-11-01	**Rev: A**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1